



Simplifying Rational Expressions: Factoring, Multiplying, and Dividing

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Learning Objectives

- Find the prime factorization of integers using a factor tree
- Simplify rational expressions by factoring numerator and denominator
- Multiply rational expressions and reduce to lowest terms
- Divide rational expressions using the reciprocal rule

Factor completely and simplify each rational expression, stating any restrictions where appropriate.

1. Warm-up: Factor the trinomial completely.

$$x^2 + 7x + 12$$

Answer: _____

2. Warm-up: Factor the difference of squares.

$$9x^2 - 25$$

Answer: _____

3. Use a factor tree to find the prime factorization.

$$72$$

Answer: _____

4. Use a factor tree to find the prime factorization.

$$180$$

Answer: _____

5. State the values of x for which the rational expression is undefined.

$$\frac{x+2}{x^2-9}$$

Answer: _____

6. Simplify the rational expression.

$$\frac{x^2 - 16}{x^2 + 8x + 16}$$

Answer: _____



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7. Simplify the rational expression.

$$\frac{2x^2 - 8}{x^2 - x - 2}$$

Answer: _____

8. Multiply the rational expressions and simplify.

$$\frac{x^2 - 1}{x + 3} \cdot \frac{x + 3}{x - 1}$$

Answer: _____

9. Multiply the rational expressions and simplify.

$$\frac{x^2 + 5x + 6}{x^2 - 4} \cdot \frac{x - 2}{x + 3}$$

Answer: _____

10. Divide the rational expressions and simplify.

$$\frac{x^2 - 9}{x^2 + 2x} \div \frac{x - 3}{x + 2}$$

Answer: _____





This worksheet mirrors the video lesson: a factoring warm-up, prime factorization via factor trees, identifying rational expressions, simplifying rationals, multiplying rationals, and dividing rationals. Problems progress from numerical factoring to polynomial operations on rational expressions.

Solutions

1. Warm-up: Factor the trinomial completely.

$$x^2 + 7x + 12$$

- Find two numbers whose product is 12 and whose sum is 7.
- The numbers 3 and 4 work since 3 times 4 equals 12 and 3 plus 4 equals 7.
- Write the trinomial as the product of two binomials.

Answer: $(x + 3)(x + 4)$

2. Warm-up: Factor the difference of squares.

$$9x^2 - 25$$

- Recognize the expression as a difference of two perfect squares.
- The square root of $9x$ squared is $3x$ and the square root of 25 is 5.
- Apply the difference of squares pattern to write the factors.

Answer: $(3x - 5)(3x + 5)$

3. Use a factor tree to find the prime factorization.

$$72$$

- Split 72 into 8 times 9.
- Break 8 into 2 times 2 times 2 and break 9 into 3 times 3.
- Collect the prime factors and write them using exponents.

Answer: $2^3 \cdot 3^2$

4. Use a factor tree to find the prime factorization.

$$180$$

- Split 180 into 18 times 10.
- Break 18 into 2 times 3 times 3 and break 10 into 2 times 5.
- Combine the prime factors and write them using exponents.

Answer: $2^2 \cdot 3^2 \cdot 5$

5. State the values of x for which the rational expression is undefined.

$$\frac{x+2}{x^2-9}$$

- Set the denominator equal to zero.
- Factor x squared minus 9 as the product of x minus 3 and x plus 3.
- Solve each factor for x to find the restricted values.

Answer: $x = 3, x = -3$



6. Simplify the rational expression.

$$\frac{x^2 - 16}{x^2 + 8x + 16}$$

→ Factor the numerator as a difference of squares into x minus 4 times x plus 4.

→ Factor the denominator as a perfect square trinomial into x plus 4 squared.

→ Cancel the common factor of x plus 4 from numerator and denominator.

Answer: $\frac{x - 4}{x + 4}$

7. Simplify the rational expression.

$$\frac{2x^2 - 8}{x^2 - x - 2}$$

→ Factor the numerator: pull out 2 to get 2 times the quantity x squared minus 4, then factor as 2 times x minus 2 times x plus 2.

→ Factor the denominator into x minus 2 times x plus 1.

→ Cancel the common factor x minus 2 to obtain the simplified expression.

Answer: $\frac{2(x - 2)}{x - 2}$ reduces to $\frac{2(x + 2)}{x + 1}$

8. Multiply the rational expressions and simplify.

$$\frac{x^2 - 1}{x + 3} \cdot \frac{x + 3}{x - 1}$$

→ Factor the numerator x squared minus 1 as x minus 1 times x plus 1.

→ Multiply across, then cancel the common factor of x plus 3 and the common factor of x minus 1.

→ The remaining factor x plus 1 is the simplified product.

Answer: $x + 1$

9. Multiply the rational expressions and simplify.

$$\frac{x^2 + 5x + 6}{x^2 - 4} \cdot \frac{x - 2}{x + 3}$$

→ Factor each polynomial: the first numerator becomes x plus 2 times x plus 3 and the first denominator becomes x minus 2 times x plus 2.

→ Write the product as one fraction and cancel the common factors x plus 2, x plus 3, and x minus 2.

→ All factors cancel, leaving a value of 1.

Answer: 1

10. Divide the rational expressions and simplify.

$$\frac{x^2 - 9}{x^2 + 2x} \div \frac{x - 3}{x + 2}$$

→ Rewrite the division as multiplication by the reciprocal of the second fraction.

→ Factor each polynomial: x squared minus 9 becomes x minus 3 times x plus 3, and x squared plus $2x$ becomes x times x plus 2.

→ Cancel the common factors x minus 3 and x plus 2 to obtain the simplified quotient.

Answer: $\frac{(x + 3)(x + 2)}{x(x + 2)}$ reduces to $\frac{x + 3}{x}$

